Multiple Disease Prediction System using ML

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Abstract

Predictive analytics in healthcare is a difficult endeavour. High-quality and sufficient data is often scarce, making it difficult to train accurate models. Identifying the most relevant features for prediction can be complex, requiring both statistical methods and domain expertise. Additionally, medical datasets are often imbalanced, with far more healthy cases than disease cases, which can skew the model's performance.

But a predictive model can eventually assist practitioners in making timely decisions regarding patients' health and treatment based on massive data. Diseases like Breast cancer, lung cancer, diabetes, and heart-related diseases are causing many deaths globally but most of these deaths are due to the lack of timely check-ups of the diseases. Other diseases like Parkinson's disease, while not typically fatal, it significantly impacts quality of life and can lead to severe disability. The above problem occurs due to a lack of medical infrastructure and a low ratio of doctors to the population. In this work, breast cancer, lung cancer, Parkinson's disease, heart, and diabetes are included. To make this work seamless and usable by the mass public, a medical test web application is made that makes predictions about various diseases using the concept of machine learning. In this work, our aim to develop a disease-predicting web app that uses the concept of machine learning-based predictions about various diseases.

Introduction

1.1 Problem Statement

Data collection and processing of that data is a tricky and worrisome task in healthcare. In this day and age of digital era, a vast quantity of multidimensional data on patients is created. The enormous, dense, and complex data must be processed and evaluated in order to extract knowledge for effective decision making. By identifying significant patterns and detecting correlations and relationships among many variables in huge databases. Diagnosis of chronic illnesses is a vital issue in the medical industry since it is based on many symptoms. It is a complex procedure that frequently leads to incorrect assumptions. When diagnosing illnesses, the clinical judgment is based mostly on the patient's symptoms as well as the physicians' knowledge and experience.

Data mining and machine learning approaches are making significant efforts to intelligently translate accessible data into valuable information in order to improve the diagnostic process's efficiency. Several studies have been conducted to explore the use of machine learning in terms of diagnostic abilities. It was discovered that, when compared to the most experienced physician, who can diagnose with 79.97% accuracy, machine learning algorithms could identify with 91.1% correctness. Machine learning techniques are explicitly used to illness datasets to extract features for optimal illness diagnosis, prediction, prevention, and therapy. Below are the diseases we will be working with in the scope of this article.

Breast cancer is a common and potentially deadly disease affecting many individuals worldwide. Early diagnosis is crucial for effective treatment and improved survival rates. Machine learning models have become instrumental in breast cancer diagnosis, leveraging large datasets of patient records and clinical data. These models can identify patterns and anomalies with high accuracy, often surpassing traditional diagnostic methods. By analyzing various factors such as menopausal status, tumor size, patient histories, etc. machine learning can help detect cancer at early stages.

Diabetesis a widespread chronic disease that affects millions of people globally, characterized by high blood sugar levels due to the body's inability to produce or effectively use insulin. Early diagnosis and management are crucial to prevent severe complications. Machine learning is used to diagnose and manage diabetes by analyzing large datasets of patient records and clinical data. These models can identify patterns and risk factors, such as glucose, blood pressure, insulin, etc. to predict the onset of diabetes.

Heart Disease is a leading cause of death worldwide, encompassing various conditions that affect the heart's function. Diagnosis using machine learning approach can help manage heart disease by analyzing large datasets of patient records and clinical data. These models can identify patterns and risk factors, such as age, cholesterol, biomarkers, etc. to predict the chances of heart disease.

Lung cancer, a serious and often fatal disease,is characterized by uncontrolled cell growth in the lungs. Machine learning is used to diagnose and manage lung cancer by analyzing large datasets of patient records, medical history, and clinical data. These models can identify patterns and risk factors, such as smoking history, anxiety, and other factors, to predict the likelihood of lung cancer.

Parkinson's disease is a progressive neurological disorder that primarily affects movement, leading to symptoms such as tremors, rigidity, and impaired coordination. While this disease shares common symptoms with several other neurological and movement disorders, which can make it challenging to predict and diagnose accurately. Symptoms like tremors, rigidity, and balance issues are not unique to Parkinson's and can overlap with those of other conditions, leading to potential misdiagnoses, so the we make predictions by training the model with a range of biomedical voice measurements.

By incorporating machine learning, healthcare providers can offer more precise and timely interventions, ultimately enhancing patient outcomes and quality of life.

1.2 Objective

The main objective of this project is to make a predictive system using the data available from that will be used to identify and diagnose diseases like breast cancer, diabetes, heart related disease, Parkinson's disease and lung cancer. The main goal is to create a predictive system using available data to identify and diagnose diseases such as breast cancer, diabetes, heart-related diseases, Parkinson's disease, and lung cancer. The project will focus on developing an interactive and user-friendly platform with React, ensuring a clean, intuitive, and responsive user interface. This interface will include forms for users to input health data like age, weight, medical history, and other relevant metrics.

Appropriate machine learning algorithms will be selected for each disease, such as logistic regression, naive bayes, decision tree classifier and random forest classifier for breast cancer, logistic regression, naive bayes and SVM for diabetes detection, logistic regression and naive bayes for heart disease, logistic regression, Naive Bayes and decision tree classifier for lung cancer, and logistic regression and SVM for Parkinson's disease. These models will be trained using large datasets from medical research databases and validated for accuracy. The trained models will then be evaluated based on their accuracy scores using training dataset and testing dataset. The model with highest accuracy score for that disease will be used to build the predictive system.

The integration and testing phase will connect the React frontend to the backend API, ensuring seamless data exchange between the user interface and the machine learning models. Thorough user testing will be conducted to ensure the system's usability, accuracy, and reliability, with feedback gathered to refine and improve the user experience. Performance optimization will be carried out to ensure the application can handle a large number of users and data inputs efficiently.

1.3 Summary

In summary, this project aims to create a powerful and accessible tool for predicting multiple diseases through a React website integrated with machine learning models. The main objective is to develop a predictive system using the data available to identify and diagnose diseases like breast cancer, diabetes, heart-related diseases, Parkinson's disease, and lung cancer. This system will provide users with valuable insights into their health, enabling early detection and proactive management of these serious conditions.In summary, this project aims to create a powerful and accessible tool for predicting multiple diseases through a React website integrated with machine learning models. The main objective is to develop a predictive system using the data available to identify and diagnose diseases such as breast cancer, diabetes, heart-related diseases, Parkinson's disease, and lung cancer. This system will provide users with valuable insights into their health, enabling early detection and proactive management of these serious conditions.

The React-based platform will be designed to be user-friendly and highly interactive, ensuring that users can easily navigate and input their health data. The system will utilize advanced machine learning algorithms to analyze this data and provide accurate predictions. By identifying potential health issues early, users can take timely action, seeking medical advice and interventions that can significantly improve their health outcomes.

Literature Survey

2.1 Related Work

A structural model and a collection of conditional probabilities are used by Bayesian classifiers. They make the assumption that the contributions of all factors are independent. It first calculates the prior probability for each class, and then applies the occurrence of each variable value to an unknown scenario. A Bayes network classifier is built on a Bayesian network, which reflects a joint probability distribution over a set of category characteristics.

The SVM method and the Nave Bayes technique were used to predict kidney disease. The authors attempted to categorize various stages of kidney disease using the suggested ANFIS algorithm. The study's purpose was to design an effective categorization algorithm using several assessment metrics such as accuracy and execution time. While the SVM Algorithm provided higher classification accuracy, the Nave Bayes fared better since it produced results in less time. The results show that SVM outperforms the Nave Bayes Approach in predicting renal illness.

The fuzzy technique with a membership function was used to forecast cardiac disease. Using the Fuzzy KNN Classifier, the authors attempted to eliminate ambiguity and uncertainty from data. The 550-record dataset was separated into 25 classes, with each class having 22 items. The dataset was separated into two equal parts: training and testing. The fuzzy KNN methodology was implemented after pre-processing techniques were used. This technique was examined using several assessment metrics such as accuracy, precision, and recall, among others. Based on the data, it was discovered that the fuzzy KNN classifier outperformed the KNN classifier in terms of accuracies.

For the prediction of cardiac disease, a novel technique based on the ANN algorithm was devised. The researchers created an interactive prediction method based on categorization using an artificial neural network algorithm and taking into account the thirteen most important clinical parameters. The suggested method proved effective for predicting heart disease with an accuracy of 80% and can be very useful for healthcare practitioners.

Authors in presented an automated approach for answer- ing difficult inquiries for heart disease prediction. The Naive Bayes methodology was used to create this intelligent system in order to provide quick, better, and more accurate outcomes. It might aid doctors in making clinical judgments about heart attacks. This system may be enhanced by including SMS functionality, building

Android and IOS mobile applications, and including a pacemaker in the order.

Diabetes and breast cancer were diagnosed by incorporating the adaptivity characteristic into support vector machines. The goal was to offer a rapid, automated, and adaptable diagnostic method using adaptive SVM. To achieve better results, the bias value in conventional SVM was changed. The suggested classifier produced output in the form of 'if-then' rules. The proposed method was used to diagnose diabetes and breast cancer, and it provided 100% right classification rates for both conditions. Future research should focus on developing more efficient ways for changing the bias value in conventional SVM.

For the prediction of type 2 diabetes, a hybrid model based on clustering followed by classification was proposed. For prediction, the suggested model uses K-means clustering and the C4.5 classification method with k-fold cross-validation. The model generated encouraging results with a classification accuracy of 88.38% using the hybrid technique, which might be highly useful for clinicians in making appropriate clinical choices related to diabetes.

Methodologies

In this framework, machine learning algorithms- support vector machine (SVM), logistic regression, random forest classifier, xgboost classifier, decision tree classifier and Naive Bayes are used. In frontend, React is used to develop a website, which is connected to python flask backend.

3.1 Machine learning

Machine learning is a subset of artificial intelligence (AI) that involves the development of algorithms and statistical models enabling computers to perform specific tasks without explicit instructions. Instead, these systems learn from and make decisions based on data. By recognizing patterns and correlations within large datasets, machine learning models can make predictions or decisions that improve over time as they are exposed to more data. Machine learning is applied in various domains like speech recognition, medical diagnosis, financial forecasting, and autonomous systems, to create solutions that can adapt and optimize their performance autonomously.

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